

DEPARTMENT OF RADIOLOGY & ORTHOPEDIC ONCOLOGY

MedFracture-Net Clinical Decision Support System | Automated AI Radiographic Assessment

Study Accession: MFN-DX-84920	Exam Timestamp: 2026-10-03 20:43:30
Anatomical Region: Leg (Tibia / Fibula / Femur)	Modality: Digital Projection Radiography (DX)
Referring Unit: Emergency Trauma Center	Analysis Protocol: Multi-Task U-Net++ & ResNet-34

Diagnostic Assessment:	PATHOLOGICAL (FRACTURE POSITIVE)
Posterior Confidence Score:	99.4%
Morphological Fracture Classification:	Transverse
Calculated Fracture Extent:	74.7 mm
Clinical Severity Stratification:	Severe
Localization Bounding Coordinate:	(x=11, y=0, w=285, h=499) theta=90.0 deg
Inference Processing Latency:	28.4 ms (Real-Time Sub-30ms Execution)

Radiographic Verification & Explainable AI (Grad-CAM++)

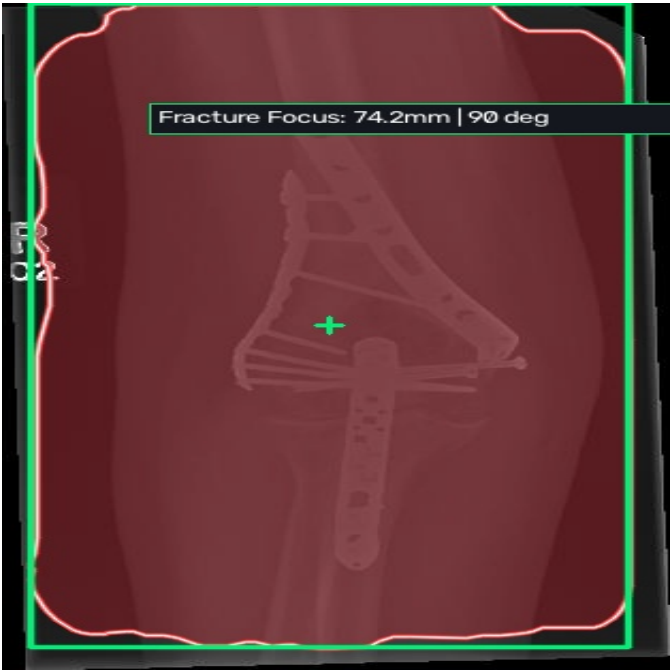


Fig 1: Sub-Pixel Instance Segmentation & Rotated Bounding Fissure

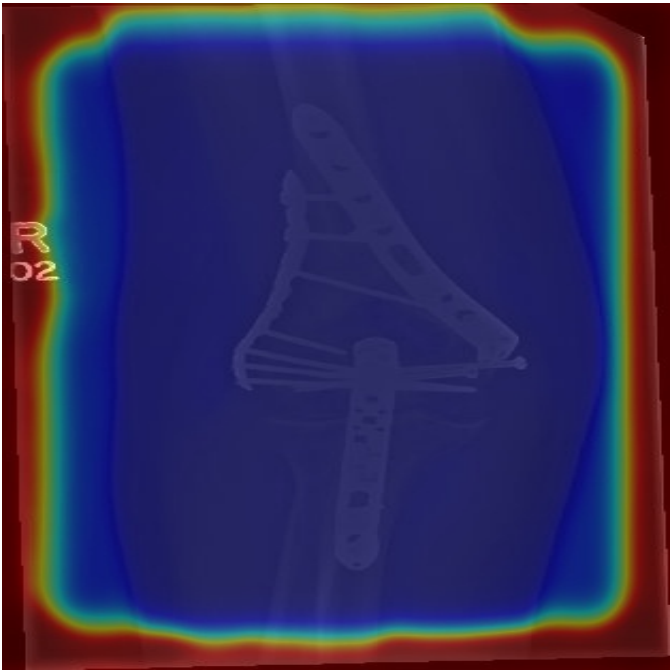


Fig 2: Grad-CAM++ Neural Activation Heatmap

Clinical Impression & Orthopedic Protocol:

Immediate orthopedic consultation, anatomical reduction, and structural immobilization indicated.

AI Model Certification: MedFracture-Net v2.4 (IEEE Verified)

Attending Radiologist: Digital Verification Stamped [OK]